Therapeutics Orals

Supervisors: Anna D (posterior eye) and Lily H (anterior eye)
N.B** No pictures were given - purely conversational
N.B*** They will stop you when there is enough information and move the convo forward
N.B**** One supervisor takes one case - they don't ask questions for other person's case
N.B*****Answers are the answers that I gave + extra info in hindsight - Use as a guide but do your own research too!

Case 1

- *A 30Y Iranian male presents to your clinic with one eye IOP 60 and the other is 17*
- *What are your ddx?*
    - AAU/PSS/AAC/FIC
- *How would you differentiate PSS vs. AAU?*
    - AAU is more painful
    - More red and inflamed
    - More cells in AC
    - Typically don't expect elevated IOP in AAU - typically lower initially due to inflamed CBB
- *How would you differentiate PSS vs. AAC?*
    - Angle is open on van herick and gonioscopy
    - AAC more painful
    - Patient profile - PSS typically young adults (male), AAC typically older (female)
    - Family history of AAC
- *Let's say angle is open and mild cells - So our tentative diagnosis is PSS. How would you mx? Specific drug name/concentration/dosage?*
    - Pred forte 1% q2h
        - In hindsight, Will's says QID
    - Timolol 0.5% (instil in office) + q mane = BID
- *How often would you review them?*
    - Review in 1 day and daily until episode has ceased + IOP back to normal
- *How long does it typically last?*
    - Hrs - days - weeks
- *Now let's say the IOP suddenly rises - What is the reason for this?*
    - Steroid response induced IOP rise
- *Who are most at risk of steroid IOP response?*
    - Armaly and Becker
        - Existing POAG
        - FHx of POAG (1st degree relative)
        - Children < 10y (Ohji and Lam)
        - Secondary: Connective tissue disorders, diabetes, myopia
- *What % of population has steroid induced IOP response?*

    - Armaly
        - < 6mmHg = No/low resposne (60%)
        - 6-15mmHg = Mod response (30%)
        - >16mmHg = Strong response (5%)
- *What would your management now be?*
    - Add topical lowering drop i.e. alpha-agonist or carbonic anhydrase inhibitor
- *What is an alpha-agonist that you would use?*
    - I said brimonidine 0.2% as i has neuroprotective effect and is on PBS
    - I'm sure you can argue apraclonidine 0.5% too
- *What are some contraindications and side effects of your chosen alpha agonist?*
    - CI: MAO inhibitors, tricyclic antidepressants, pregnancy, children, CNS issues?
    - Follicular conjunctivitis, hyperaemia, stinging

Case 2

- *Let's say you diagnose a case of AAU - what other tests need to be performed?*
    - NaFl assessment to rule out epithelial defect
    - Assess AC for cells/flare in 1mm x 1mm conical beam
    - Gonioscopy to check angle for PAS
    - Check pupils/iris for posterior synechiae and nodules
    - IOP to rule out PSS and for baseline if steroid response occurs
    - Check lens for any PSC (as will use steroid)
    - DFE for intermediate/posterior involvement
- *What history questions would you ask?*
    - Systemic work up
        - Rheumatological: spondyloarthropathies, especially ankylosing spondylitis
        - Genito-urinary: Reiter's syndrome, Behçet's disease, syphilis
        - Respiratory: sarcoidosis, tuberculosis
        - Gastrointestinal: inflammatory bowel disease
        - Dermatological: psoriasis, syphilis, sarcoidosis, herpes simplex, herpes zoster, Behçet's disease.
- *Would you suspect everyone with back/joint problems? Is there a particular feature that would warrant further investigation?*
    - When their is pain without stimulus i.e not after exercise but just from sitting
    - When pain is worse in the morning
- *How would you manage this patient?*
    - Steroid
        - Pred forte 1% q2h or q1h based on severity and loading dose q15m for 1hour
    - Cycloplegia
        - Atropine 1% tid
            - Photophobia
            - Stabilise blood aqeuous barrier
            - Prevent posterior synechiae

- *If you note posterior synechiae, how do you break it?*
    - Start with atropine 1% → 2.5% phenylephrine → 10% phenylephrine → cotton pledget soaked in 10% phenylephrine
- *When would you send to GP for systemic work up?*
    - Bilateral
    - Systemic work up reveals indication
    - Granulomatous
    - Recurrent
    - Chronic
- *When would you send to GP vs. ophthalmologist?*
    - Presence of hypopyon
    - Posterior involvement
    - Not responding to treatment

Case 1 - herpes simplex epithelial keratitis

Given picture of dendritic ulcer - describe what you see, what other signs and symptoms would you expect

Symptoms - tearing, photophobia, pain

Signs - skin vesicles, secondary uveitis with KPs, borders stain with rose bengal/lissamine green

Ddx

- Acanthamoeba keratitis
- Herpes zoster keratitis
- Healing RCE/corneal abrasion
- Bacterial keratitis

Explain pathophysiological of herpes simplex vs herpes zoster

Would there be a differing presentation if it was HSV1 vs HSV2?

What differentiating signs would you see in zoster

Treatment and review period, how long would you expect this to last? (remember epithelial debridement!)

Case 2 - pseudoexfoliation

Given picture of pupil - what do you see, what other signs do you expect to see

What other tests would you do (glaucoma workup)

What is the profile of a PXF patient (Caucasian)

If IOPs were 15mmHg in the affected eye and 14 in the other eye what would you do? (diurnal IOP) If morning IOPs were 14 and 17 what would you do? (review closely due to risk of glaucoma - EMGT)

Review period? (every 3 months)

Other risk factors for glaucoma

If at 2nd/3rd review IOPs are 14 and 20 what would you do? (start treatment)
What drug would you choose? (PGA) Contraindications/side effects
If PGAs contraindicated what would you choose? (timolol 0.5%) Contraindications/side effects
Why would you choose timolol 0.5% over 0.25% (higher efficacy, darker irises)

Case 1: MK

shows white light photo of epithelial defect close to limbus, irregular shape but looks relatively small….

What do you see?

- Possible epithelial defect with infiltrates

What are you DDx?

- Marginal keratitis and microbial keratitis

How can you tell the difference between microbial keratitis and marginal keratitis?

- Size, location, shape, AC reaction, discharge, lid oedema

Risk factors for bacterial keratitis?

- Contact lens wear, trauma, ocular surface disease

Common causative organisms for bacterial keratitis in CL wearer and non CL wearer?

- CL: pseudomonas aeruginosa
- non-CL: Staph aureus

How would you treat this?

- Monotherapy with ciprofloxacin 0.3%
- Q 15 mins for 6 hrs
- Q 30 mins for rest of day
- Q 1 hr overnight and next day
- Review next day

What do you expect to see when you review them?

- Possible mild symptoms improving: pain
- Not getting worse in terms of signs: size of ulcer, etc

What else would you give?

- Cycloplegia atropine 1% if there is AC reaction

Would you consider steroids?

- Not in this case because it's peripheral
- SCUT showed that may be useful if worse VA on presentation and central ulcer
- If i did do a steroid, i would do it after wound has healed

What's another treatment option?

- Dual therapy: compounded: tobramycin 1.3% and cephazolin 5%

When would you consider using dual therapy?

- If large deep ulcers in elderly due to risk of corneal perforation
- If potentially sight threatening - 1,2,3 rule

Why does there need to be a combination of the two drugs and mechanism of action of drugs?

- Cephazolin is beta lactam and inhibits cell wall synthesis - targets gram positive bacteria
- Tobramycin is protein synthesis inhibitor and has more gram-negative (pseudomonas) coverage
- Synergistic effects if used one after the otherd

<u>Case 2: NTG: 30 y.o patient presents for a routine eye exam</u>

Shows photo of two nerves: asymmetric in size and asymmetric C/D

What's important to assess when looking at discs (glaucoma related)?

- Don't forget ISNT rule!!!!
- Colour, disc haemorrhages, lamina cribosa, RNFL, size, PPA

Does this nerve look healthy?

- Doesn't follow ISNT rule (Superior rim is thinner than nasal)

What sort of VF defect do you expect to see?

- Inferior arcuate or nasal step

What sort of VF defects do you expect for NTG?

- More central and close to fixation

What other tests might you do?

- CCT, IOP, gonio, retinal photos, VF, stereo disc photos, OCT of RNFL and GCA

IOP is R12, L12, what else would you do?

- Phased IOP

shows photo of VF: clover-leaf and high false negatives (67%)

This is the VF, describe it:

- Good fixation, poor false negatives - unreliable
- Clover-leaf pattern - likely to be due to inattention or malingering

What would you do?

- Repeat VF

If you repeated VF and there's no defect what would you do?

- See them in a year for routine review and repeat VF

If you repeated VF and there was a defect, what would you do?

- Make sure it's repeatable - VF in 3 months

If the defect is repeatable and this is a 70 y.o px would you treat and what would you treat with?

- Yes, PGA xalatan 0.005% nocte

What are the contraindications of PGA?

- CMO, uveitis and recurrent HSK

Let's say this px has CMO, what would you give?

- Beta blockers timolol 0.5% mane

Why 0.5%?

- (thanks Anna Y): better drug efficacy and for dark irides

If the px was 30 y.o would you treat?

- Yes!!!!!!!!!!! → more likely to impact central vision

Are you definite that it's NTG?

- Must rule out neurological problems

Joseph

PXF – photo of lens

- What do you see?
- What else would you usually see in this syndrome?
- Patient profile?
- How does this differ from PDS in terms of signs and patient profile?
- What is your work up?
- Would you treat?
- What IOP would you consider treatment?
- If it was 10mmHg difference would you treat?
    - What drops
        - § PGA contraindications
    - What if PGA contraindicated
        - § Beta blockers contraindications
    - What drops would you choose in pregnancy?
        - § What class is bimonidine
- When do you review?
- If no glaucoma when do you review?

MK – marginal vs bacterial

- What is your diagnosis?
- How can you differentiate between microbial and marginal keratitis?
- How would you treat bacterial?
- When do you refer?
    - Why do you refer old people?
- What is the dual therapy? And how they work?
- What the difference between gram positive and gram negative?

you’re a god mate
Anna just showed me disc photos and asked me to describe. Don't mention size coz you can't tell without fovea in shot (which is what i did lol) and then said c/d, there was a splinter haem inferiorly in one eye. Then she gave me age, iop, cct and fields were clear
the cct were 585 which i said was thicker end of normal and then she asked for range, i said 510-580 with a mean of 550. she was like thats a big range where are you getting that from. i said nhmrc but not sure if right or not

then chose a PGA she asked contraindications dug myself a hole with pregnancy coz i said only in last trimester its bad. she was like which ones can you use, instead PGAs. i was like no coz then you would have to change drugs during pregnancy. she asked what category i was like b3 i think
then she asked next choice i said beta blocker then she asked contraindications. i said asthma, COPD, heart problems, systemic beta blockers, diabetics due to masking hypoglycaema. then she asked what else
she also asked specifically what heat condition. but at this point she was like youre doing really well. Also i forgot what C stands for in COPD, i said coronary but its chronic

Lily just gives you a photo and asks what is the most likely diagnosis? Mine was Uveitis
and then what is your workup? this includes history questions and clinical shit like what you would look for like corneal signs. i mention NaFl to exclude bacterial keratitis. i said evert lids. I said do dfe to check for intermediate and posterior involvement and said i’d use atropine (to break synechiae, support BAB, reduce ciliary spasm etc). Then use phenylephrine if syneichiae not broken. She asked what is good about using atropine and phenylephrine work well together. Why not atropine and cyclo together. What would you do if atropine and phenyl didnt break the synechiae what would you do?
When would you refer? I said systemic cause, chronic, recurrent, herpes, not responding to treatment, hypopyon, posterior involvement. But she wanted something more...

Anna: Picture of optic nerve with L disc haemorrhage

1. Describe the disc (no fovea so cant say how big it is, NRR, PPA, disc haemorrhage, difficult to see NFL)
2. DDX: NTG, POAG, PVD, diabetes, HTN, physiological cupping (but haem)
3. Then revealed information: 26 IOP OU, CCT 585, FHx mum glaucoma, VF clear
4. Work up: IOP phasing, gonio, repeat VF for baseline, OCT, DFE (see if PVD but px asymptomatic so unlikely)
5. DX: OHT high risk of glaucoma due to FHx, high IOP, and if drance haem > risk for having glaucoma and progression of glaucoma. Could also be pre-perimetric glaucoma, overestimated IOP (Moorefields vs. NHMRC)
6. Management: make sure not PVD, GP for work up (HTN, DM), OCT in 3 months to see if RNFL loss. If a defect tx because tx reduces risk of developing glaucoma in 50% OHT
7. If you were to treat, what would you treat with?
    a. PGA: latanoprost 0.005% at night. contraindication? onset of effect, max effect, IOP reduction
    b. BB: timolol 0.25%. Why? Because most effective for light coloured eyes. Why in the morning? Because IOP is highest in the morning due to aq production. Not as effective at night because not as much aq production. I mentioned something about hypotension at night which i retracted… LOL contraindication, onset of effect
8. Steroid response- what percent of population responds, who is most likely to, when does it happen

Lily: Picture of eye with synechiae

1. Give a diagnosis. AAU
2. Work up: Cells and flare, fluorescein, gonio, IOP, DFE, 5 B's
3. MX: break synechiae. With atropine, if that doesnt work what next? Phenylephrine. Why does that work, because different mechanism of action. If that doesnt work, pledge- soak cotton with phenylephrine 10 and put in fornix for 15min?? moderate: predforte q1h until 2 step improvement in SUN, qid until zero and taper dropping a drop each week. Atropine 2X day. What are the contraindications for atropine: I said narrow angles, old people with cardiovascular problems
4. When would you refer: not improving or responding to tx, systemic assoc., posterior involvement, hypopyon, chronic, recurrent, and i should have said bilateral